Exit Protocol

Virtual Escape Room Experience

Software Requirements Specification

CSCE 247

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1. Introduction

Purpose:

With the world of true crime and puzzle-solving apps, and mystery podcasts seeing a rise in popularity,, the demand and market potential for an engaging and accessible escape room experience is higher than ever. Finding the perfect website that's clean, and easily accessible is difficult among the sea of other sites, our site will sleekly allow users to explore different escape room options, compare features like difficulty level, themes (mystery, horror, fantasy, historical), and match players with others who share their interests. Whether someone is a die-hard true crime enthusiast, a casual gamer, or part of a team-building group, this platform will make finding the perfect escape room seamless and enjoyable.

Source: https://www.edisonresearch.com/the-true-crime-consumer-report-by-edison-research-and-audiochuck/

2. Stakeholders

- True Crime Enthusiants
- Casual Gamers
- Groups
    - Potential Corporate/General Team building
    - Friends and Social Group Activity
- Puzzle/Logic Game Fans

Personas:

JOSEPH COOK

BIO:

Joseph Cook is a single archaeologist living in Miami, Florida. A lifelong puzzle enthusiast, thrive on brain teasers and challenging discoveries. They avoid tedious tasks, preferring mysteries that spark curiosity and insight. Always ready to face the unknown, sees archaeology as the ultimate puzzle.

AGE	25
STATUS	Single
LOCATION	Miami, Florida
OCUPATION	Archaeologist
ARCHETYPE	Puzzle Enthusiast

PERSONALITY

Extrovert — Introvert

Sensing — Intuition

Judging — Perceiving

FUSTRATIONS

- Many escape room games in Miami are costly, making it hard to play as often as he'd like.
- Online or virtual escape rooms don't feel as immersive, leaving him wishing for more creative digital options when he can't attend in person.
- Some escape rooms are too repetitive or gimmicky, which frustrates him because he craves clever, stimulating puzzles rather than tedious tasks.

INTERESTS

- Escape Room Games – enjoys the challenge, teamwork, and puzzle-solving aspects, though he's often frustrated by access and cost.
- Archaeological Discovery – Treats fieldwork and artifact analysis like real-world puzzles that connect to history.
- Brain Teasers & Logic Games – From riddles to mechanical puzzles.

GOALS

- CHALLENGE HIS PROBLEM-SOLVING SKILLS IN A FUN, HIGH-PRESSURE ENVIRONMENT THAT MIRRORS THE PUZZLES HE LOVES IN ARCHAEOLOGY.
- STRENGTHEN TEAMWORK BY COLLABORATING WITH FRIENDS OR COLLEAGUES TO SOLVE MYSTERIES TOGETHER.
- IMMERSE HIMSELF IN A LIVE PUZZLE EXPERIENCE, SATISFYING HIS CRAVING FOR INTERACTIVE, HANDS-ON DISCOVERY.

MOTIVATIONS

- CURIOSITY FOR THE UNKNOWN – HE'S DRIVEN BY THE THRILL OF SOLVING MYSTERIES, WHETHER ANCIENT RUINS OR MODERN PUZZLES.
- LOVE OF MENTAL CHALLENGE – HE SEEKS OUT ACTIVITIES THAT KEEP HIS MIND SHARP AND ENGAGED, AVOIDING ANYTHING THAT FEELS DULL OR REPETITIVE.
- DESIRE FOR CONNECTION – EVEN THOUGH HE'S SINGLE, HE VALUES OPPORTUNITIES (LIKE ESCAPE ROOMS OR PUZZLES WITH FRIENDS) TO COLLABORATE AND SHARE DISCOVERIES.
- PASSION FOR DISCOVERY – EACH SOLVED PUZZLE, WHETHER IN ARCHAEOLOGY OR GAMES, FUELS HIS SENSE OF ACCOMPLISHMENT AND PURPOSE.

EMILY CARTER

BIO:

A 34-YEAR-OLD MARKETING MANAGER FROM CHARLOTTE, NC, WHO LOVES MYSTERIES, PUZZLES, AND SOCIAL GAMING WITH FRIENDS. THEY SEEK FUN, IMMERSIVE ESCAPE ROOM EXPERIENCES BUT GET FRUSTRATED BY CLUTTERED WEBSITES AND UNCLEAR OPTIONS.

AGE	34
STATUS	Married
LOCATION	Charlotte,NC
OCUPATION	Marketing manager
ARCHETYPE	True Crime lover

PERSONALITY

Introvert — Extrovert

Intuition — Sensing

Judging — Perceiving

GOALS

- Enjoy immersive escape room experiences – more than just a simple game; they seek challenging, story-driven experiences that feel like stepping into a mystery.
- Connect with friends and coworkers virtually – value activities that allow them to maintain strong social bonds, even when distance or busy schedules keep them apart.
- Combine fun with true crime themes – aim to find games that blend their love of puzzles with the intrigue of investigative stories, making the experience both entertaining and mentally stimulating.

FUSTRATIONS

- Cluttered or confusing websites – Struggles with poorly organized information that makes planning activities difficult.
- Difficulty comparing escape rooms – Wants straightforward, side-by-side comparisons to make informed choices quickly.
- Limited time or accessibility – Frustrated when logistical issues like scheduling or location make participation harder.

INTERESTS

- True crime podcasts & mystery shows – They enjoy diving into crime stories and investigative series, drawn to the thrill of piecing together clues and uncovering hidden details.
- Board games, trivia nights, casual gaming – They love interactive, competitive activities that challenge their knowledge and problem-solving skills while keeping things lighthearted and social.
- Virtual activities with friends & coworkers – Staying connected online is important, so they seek engaging virtual games and events that bring people together across different locations.

MOTIVATIONS

- Passionate about puzzles and mysteries – Enjoys solving challenges, uncovering hidden details, and engaging in mentally stimulating activities.
- Seeks fun and social experiences – Thrives on interactive activities that connect them with friends, coworkers, and like-minded people.
- Motivated by engagement and discovery – Looks for experiences that are not only entertaining but also mentally rewarding and memorable.

ALLY PAGE

BIO:

ALLY IS A JUNIOR STUDYING DIGITAL MEDIA AT HER UNIVERSITY. SHE HAS ALWAYS BEEN PASSIONATE ABOUT VIDEO GAMES AND STORYTELLING, WHICH INSPIRED HER TO PURSUE A CAREER IN MEDIA DESIGN. IN HER FREE TIME, ALLY SPENDS HOURS ON ONLINE MULTIPLAYER GAMES WITH HER FRIENDS ACROSS THE COUNTRY. SINCE SHE AND HER FRIENDS LIVE OFF CAMPUS AND DON'T OWN CARS, MOST OF THEIR HANGOUTS HAPPEN VIRTUALLY. ALLY ENJOYS CONNECTING WITH HER FRIENDS THROUGH ONLINE ESCAPE ROOMS, PUZZLE GAMES, AND FANTASY ADVENTURES. SHE'S BUDGET-CONSCIOUS BUT VALUES EXPERIENCES THAT LET HER FEEL CONNECTED AND ENGAGED.

AGE	20
STATUS	Single
LOCATION	Norfolk ,Va
OCUPATION	Student
ARCHETYPE	Creative Thinker

PERSONALITY

Extrovert — Introvert

Sensing — Intuition

Perceiving — Judging

GOALS

- Looking for fun and affordability: Ally wants an online escape room that won't break the bank but still delivers an enjoyable and exciting experience for her and her friends.
- Seeks engaging and immersive gameplay: She values a game that pulls players into the story with challenging puzzles and a captivating atmosphere, making everyone feel connected even when apart.
- Wants a social, remote-friendly experience: Ally's ideal escape room lets her and her group play together seamlessly from different locations, perfect for casual weekends or game nights.

FUSTRATIONS

- Limited budget for outings: Ally can't afford extra costs for in-person activities or traveling, so she looks for affordable or free options.
- Friends scattered and no easy transport: Her friends live far away and without cars, making it tough to meet up in person.
- Feels left out of social events: Ally dislikes missing out on fun group activities that require everyone to be physically together

INTERESTS

- Finding affordable ways to connect: Ally is motivated to discover budget-friendly activities that allow her to have fun and stay social.
- Maintaining friendships despite distance: She wants to keep her friendships strong by finding ways to engage and spend time with friends remotely.
- Engaging in creative, immersive experiences: Ally seeks interactive and mentally stimulating games or activities that captivate her and her group, especially ones that can be played online.

MOTIVATIONS

- Limited budget restricts social options: Ally feels frustrated by not having enough money for in-person activities or travel, which limits her ability to hang out with friends.
- Distance and lack of transportation: She struggles with friends living far away and no easy way to meet up due to lack of cars or convenient transport.
- Missing out on physical social events: Ally dislikes feeling left out of social activities that require everyone to be in the same place.

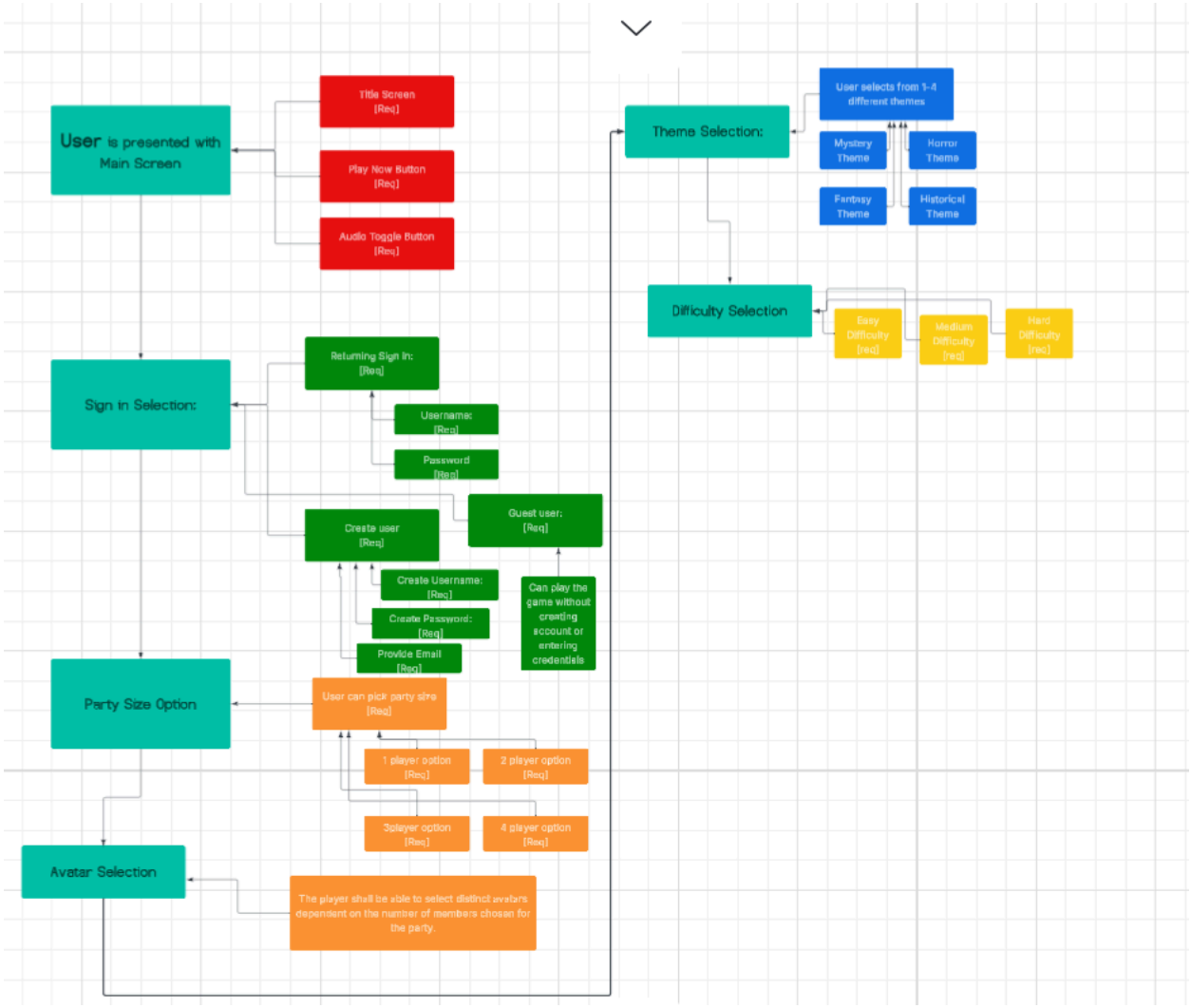
3. Constraints

- *Scheduling Constraints*
    - Simonizing schedules across group members could potentially serve as an obstacle going forward.
- *Technical Constraints*
    - Project will be written in java and accessible across different browsers
- *User and community growth*
    - While there exists a group of potentially excited customers, retention of this clientele might be difficult.

4. Overall Description

The virtual escape room system will run as a web-based and mobile-accessible application written in Java. The backend will handle game logic, databases, and connections to other services, while the frontend will run in standard web browsers or mobile devices. The user will be able to select the style(mystery, fantasy, horror, or historical) and corresponding style components/design elements will be generated alongside the respective escape puzzle. The system will connect to a database for storing user accounts, escape room details, and progress.

5. Business Use Cases



6. Functional Requirements

https://docs.google.com/spreadsheets/d/1x2JvpbzexpXUAVN_0YaDnJbXTLPJxVldrIxUEP0j_XI/edit?usp=sharing

7. Non-Functional Requirements

- Look and Feel Requirements
    - The Interface should be easily accessible for first time users

    - The Interface have high general ease of use
- Usability Requirements
    - The system shall be able to used by singular of multiple parties on the same device
- Performance Requirements
    - The system shall be able to be run by any computer regardless of hardware specifications
- Maintainability and Support Requirements
    - The system shall be able to run through any system that can run Java
- Security Requirements
    - The user's information shall be stored locally
- Legal Requirements
    - The system shall comply with all state and local law

8. Definitions and Acronyms

Java – A high level programming language used for developing a lot of different applications

9. Competitive Analysis

Product / Features	Online-Escape-Room.com	Breakout EDU	Enchambered (Alone Together)	The Hogwarts Escape
Device/Browser Support	Chromium, iOS/Android (via web browser), Edge, Safari, Firefox	Chromium, Edge, Safari, Firefox, iOS/Android[4]	Chromium, Edge, Safari, Firefox, iOS/Android (via web browser)	Chromium, Edge, Safari, Firefox (via PC)
Format	Website-based (a unique link is generated for each session for players to use to join)	Website-based, mobile app-based	Website-based	Website-based
Supports Native Multiplayer?[1]	Yes	No	Yes (players visit separate web pages)	No
Timed?	Yes	Yes	No	No
Unique Roles?[2] (for apps supporting native multiplayer)	Yes (optional via the "Free Play" multiplayer mode)	N/A	Yes	N/A
Supports In-App Communication? (for apps supporting native multiplayer)	Yes	N/A	No	N/A
Provides Hints?[3]	Yes	Limited (educator provides hints)	No	No
Positive Analysis Notes	- Includes introductory video for both the session leader and other players - Variable difficulty selection appeals to players of all ages - "Streaming" and "Free Play" game modes allow for both synchronous	- User-generated content creates variety in challenges	- Aesthetically pleasing, featuring high-quality images and sound effects	

	and asynchronous gameplay			
Negative Analysis Notes	- Limited choice of escape rooms	- Designed for educational use; lacks mainstream appeal - Relies on user-generated content, so quality varies	- Interactive elements are unclear	- Appeals to fans of the Harry Potter series; lacks mainstream appeal - Utilizes static web pages with images; lacks significant user interaction

[1] "Native Multiplayer" means that there is multiplayer functionality integrated within the application itself. Applications designed for solo use which can be used in a multiplayer fashion (i.e. by a group of players viewing the same device) are considered to lack native multiplayer.

[2] "Unique Roles" means that each player has a different role and sees a different screen. This is applicable only for apps with native multiplayer.

[3] Hints as to the solution of a riddle may be provided to players upon request if they are stuck on or do not wish to complete a particular riddle. Clues given to players upon completion of a riddle are not considered hints.

[4] Mobile app only supports mobile devices which allow for RFID/QR code scanning.

Product / Features	**Trace**	**A Dark Room**	**Spy Apprentice**	**Daily Room Escape**
Device/Browser Support	Chromium, Edge, Safari, Firefox	Chromium, Edge, Safari, Firefox, Nintendo Switch	Chromium, Edge, Safari, Firefox, iOS/Android (via web browser)	Chromium, Edge, Safari, Firefox
Format	Website-based	Website-based, app-based	Website-based; via Google Forms	Website-based
Supports Native Multiplayer?	No	No	No	No
Timed?	No	No	No	No
Unique Roles? (for apps supporting native multiplayer)	N/A	N/A	No	N/A
Supports In-App Communication? (for apps supporting native multiplayer)	N/A	N/A	No	N/A
Provides Hints?	No	No	Yes	Yes
Positive Analysis Notes	- Aesthetically pleasing and visually engaging, with high-quality graphics	- Features complex, engaging narrative	- Clear, linear path	- Aesthetically pleasing - Unique challenges each day enhance replay value
Negative Analysis Notes	- Lacks significant replay value	- Lacks images or other visual reinforcement; may not appeal to players seeking a more interactive experience	- Google Forms format limits uniqueness and interactive elements	- Lacks clear underlying theme or narrative

Summary

Existing escape room platforms and applications generally succeed in providing engaging narratives, accessibility across a range of devices and browsers, and even some innovative features such as unique roles. However, most contain limitations such as lack of interactivity, lack of mainstream appeal, limited replay value, and inconsistent content quality. As such, there is a clear opportunity to create an escape room application with broad appeal, interactivity, and narrative depth.

10. References

https://aws.amazon.com/what-is/java - What is Java?

11. Appendices